chp no 4

Measures of Dispersion

Dispersion: By dispersion we mean the extant to which the values are spread out from average.

Types of measures of Dispersion.

There are two types of measures of dispersion

1. Absolute Measures of Dispersion.
2. Relative Measures of Dispersion

Absolute Measures of dispersion: The absolute measure of dispersion measures the variation present among the observations in units of the variable.

The following measures of absolute dispersion are in Common use

1. Range 2. Quartile Deviation 3. Mean or Average Deviation.
4. Standard Deviation and Variance

Relative Measure of Dispersion: The relative measure of dispersion measures the variation present among the observations relative to their average. It is independent of units of measurement It is used for Comparison of data measured in different units

The measures of relative dispersion are

I. Coefficient of Range
II. Coefficient of quartile Deviation.
III. Coefficient of Average Deviation
IV. Coefficient of variation

Qualities (Properties) of Good Measure of Dispersion.

1. It should be simple to understand and easy to Calculate
2. It should be clearly defined by a mathematical formulas
3. It should be based on all the observations of data.
4. It should be less affected by extreme values
5. It should be capable of further mathematical treatment.

Range: Range is the difference between the largest and smallest values in the data

Range = $X_m - X_o$

Where

X_m = largest value in ungrouped data

X_o = smallest value in ungrouped data

For grouped data

X_m = upper class boundry of highest class or class mark of highest class

X_o = lower class boundry of lowest class or class mark of lowest class

Coefficient of Range = $(X_m - X_o) / (X_m + X_o)$

Merits of Range

1. It is simple to understand and easy to calculate
2. It is used in quality Control, daily temperature, stock prices etc
3. It saves Computational time

Demerits of Range

1. It is not based on all the observations
2. It depends upon extreme values only
3. It is not Capable of further mathematical treatment

Quartile Deviation

It is half of the difference between third and first quartile

$Q_D = (Q_3 - Q_1) / 2$

Coefficient of $Q_D = (Q_3 - Q_1) / (Q_3 + Q_1)$

Merits

1. It is simple to understand and easy to calculate.
2. It is not affected by extremely large or small observations

Demerits

1. It gives no information about position of observation lying outside the two quartiles.
2. It is not amendable to mathematical treatment

Mean Deviation / Average Deviation

The mean deviation is an average of the absolute deviations measured from some central points that are (mean, median, mode).

$MD_{(\bar{x})} = \Sigma|X-\bar{x}| / n$

$MD_{(Median)} = \Sigma|X-Median| / n$ $MD_{(Mode)} = \Sigma|X-Mode| / n$

Co-efficient of $MD_{(\bar{x})} = MD / X$

Merits

1. It is simple to understand and easy to calculate
2. It is based on all the observations.

Demerits

1. A is not amenable to mathematical treatment
2. It is an unsatisfactory for statistical inference
3. It is affected by extreme values.

Standard Deviation

Standard Deviation is positive square root of Arithmetic Mean of squared deviations of observations measured from Arithmetic Mean.

$$SD = \sqrt{\Sigma(X-\bar{x})^2 / n}$$

Variance Variance is mean of squared deviations of the observations from their mean.

$$Variance = \Sigma(X-\bar{x})^2 / n$$

Coefficient of Variation. Coefficient of Variation expresses the standard deviation as a percentage of the arithmetic mean. Coefficient of variation is independent of units employed. It is used to Compare the dispersion of two or more sets of data that are measured in different units. It is also used as a criterion of Consistent performance, the smaller the coefficient of Variation is considered to be more consistent in the performance.

$$C.V = (SD/\bar{X}) \times 100$$

Properties of Variance.

1. Variance of a Constant is equal to zero that is Var (a) = 0 where 'a' is constant
2. Variance is independent of origin. that is Var ($x \pm a$) = Var (x)
3. Variance is not independent of scale that is Var (ax) = a^2 Var (x)
4. Variance of sum or difference of two independent variables is equal to sum of their respective Variances that is

Var ($X \pm Y$) = Var(X) +Var(y) where x and y are independent variables.

5. If n1 observations have mean $\bar{x}_1$ and variance $S1^2$

   n2 observations have mean $\bar{x}_2$ and variance S_2^2,and so on

   If nk observations have mean $\bar{x}_k$ and variance S_K^2.

Then Combined variance of total observations is

$$S^2c = \Sigma n_i \,[S^2_i(\bar{x}_i - \bar{x}_c)^2] / \Sigma n_i$$

6. For normal distribution

   68.27% of the values ties between $\bar{X} \pm S$

   95.45% of the values lies between $\bar{X}+2S$

   99.73% of the values lies between $\bar{X} \pm 3S$

Symmetrical Distribution: A distribution is said to be symmetrical if the values equidistant from the central have same frequencies.

maximum

For symmetrical distribution Mean = Median = Mode

Skewness. The departure from symmetry in a distribution is known as skewness.

Positively Skewed Distribution. A distribution is said to be positively Skewed if right of the Curve is longer than left tail.

For positively skewed distribution: Mean > Median > Mode.

Negatively Shewed Distribution: A distribution is said to be negatively Skewed if left tail of the Curve is longer than right tail

For negatively showed distribution: Mode > Median Mean.

Kurtosis: Kurtosis is the degree of peakedness of the Curve.

Moment: Moments are arithmatic mean of the powers to which the deviations are raised.